

Inheritance Data: Predict, Calculate, Evaluate

I can calculate inheritance probabilities, ratios and percentages and evaluate why observed results can differ from predictions.

PREDICT

from the square

CALCULATE

$\text{frequency} \div \text{total} \times 100$

EVALUATE

chance and sample size

[Open the original interactive lesson](#)

What success looks like

- I can convert Punnett outcomes into fractions, ratios and percentages.
- I can calculate observed percentages from a dataset.
- I can compare observed and predicted outcomes and explain the role of chance and sample size.

Retrieve and reconnect

- State the phenotype ratio for $Pp \times Pp$.
- Convert $1/4$ to a percentage.
- Calculate the total of 31 and 9.
- Lead-in: distinguish expected from guaranteed.

A $Pp \times Pp$ cross produced 31 purple and 9 white seedlings. Which first calculation enables comparison with 75% : 25%?

Think first. Point, say, sign or write your choice. Give one reason.

A. Calculate each frequency $\div 40 \times 100$.

B. Subtract 9 from 31 and stop.

C. Declare the Punnett square wrong because the counts are not exactly 30 and 10.

Look closely · what is the evidence?

EXPECTED · model probability

OBSERVED · recorded frequency

COMPARE · same unit, cautious conclusion

Look closely · what is the evidence?

Pp × Pp → 3 purple : 1 white

40 seedlings: $0.75 \times 40 = 30$ purple, $0.25 \times 40 = 10$ white expected

Expected

observed: 31 purple, 9 white

$31 \div 40 \times 100 = 77.5\%$ · $9 \div 40 \times 100 = 22.5\%$

Observed

77.5% vs 75% expected — consistent with the model

chance decides each seed: a bigger sample usually lands closer to 3:1

Compare

Words we will use

- probability — chance of an outcome
- ratio — relative amounts
- sample size — number observed

Watch the model

- From $Pp \times Pp$, predict $3/4$ purple and $1/4$ white: 75% and 25%.
- For 40 seedlings, calculate expected counts: $0.75 \times 40 = 30$ purple; $0.25 \times 40 = 10$ white.
- Calculate observed percentages: $31 \div 40 \times 100 = 77.5\%$; $9 \div 40 \times 100 = 22.5\%$.

Which comparison is scientifically correct?

A. Observed 77.5% purple is close to predicted 75%; chance can explain a small difference.

B. Any difference proves the Punnett model is false.

C. A 3:1 prediction guarantees every batch will be exactly 3:1.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Use repeated batches: individual groups can vary more than the combined total.
- A larger sample often gives an observed proportion closer to the expected probability, although exact agreement is not guaranteed.
- Evaluate the model: useful for a simple monohybrid cross, limited for complex traits or unequal gamete/offspring survival.

Find and repair the misconception

The observed ratio is not exactly 3:1, so the Punnett square must be wrong.

A. Keep it: probability must match every sample exactly.

B. Repair it: chance causes variation; compare percentages and consider sample size.

C. Repair it: change one observed seedling so the data fit the model.

Expected versus observed data lab

Classify predictions, observations and valid conclusions, then repair the guarantee claim.

- PREDICT
- COUNT
- PERCENT
- COMPARE
- EVALUATE

Expected versus observed data lab

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- EXPECTED FROM MODEL
- OBSERVED IN DATA
- CAUTIOUS CONCLUSION

State the difference, explain why it can occur and name one model limitation.

Your independent mission

Complete a full inheritance data response: Punnett prediction, ratios, probabilities, observed percentages and evaluation.

Record: Visible working from cross to percentages, a correct expected-observed comparison and an evaluation naming chance, sample size and one model limit.

Choose your support and challenge

- Supported: Complete a partly filled square, convert $\frac{3}{4}$ and $\frac{1}{4}$ to percentages, then choose the valid data conclusion.
- Standard: Build the square, calculate expected counts and observed percentages, then explain the small difference.
- Stretch: Analyse repeated batches and the combined sample, then evaluate sample size and the monohybrid model's limits.

Show what you know

Compare 77.5% observed with 75% predicted in one cautious sentence.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.